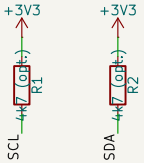
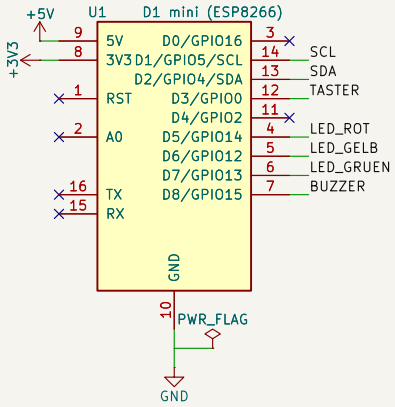
CO2-Ampel mit Wemos D1 mini (ESP8266)
- CO2 + Temperatur + Luftfeuchte ueber I2C-Sensor
- Anzeige auf OLED, Ampel-LEDs, Summer, Taster
- WLAN ueber ESP8266

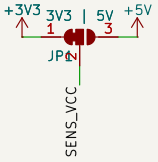


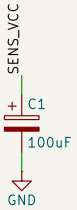
I2C-Pull-ups: nur bestuecken, wenn weder
Display noch Sensor eigene Pull-ups haben.



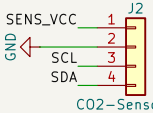
OLED-Display 128x64, I2C (Adresse 0x3C)
Pinfolge GND-VCC-SCL-SDA vor Bestellung pruefen!







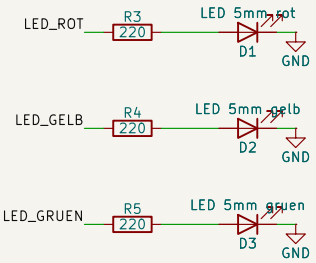
JP1: Sensor-Versorgung waehlen
1-2 = 3,3 V (Standard), 2-3 = 5 V

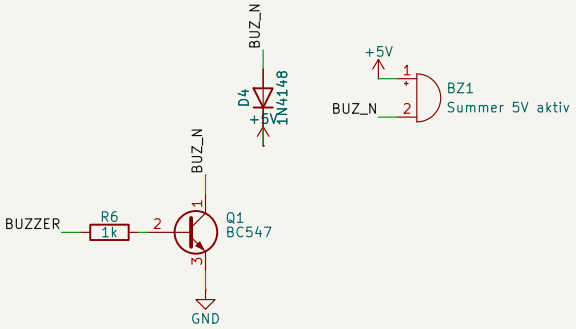


CO2-Sensor (z.B. SCD40/SCD41/SCD30), I2C
Pinfolge VCC-GND-SCL-SDA an Sensormodul anpassen!



Taster an D3/GPIO0 (interner Pull-up).
Beim Einschalten nicht druecken (Flash-Modus).

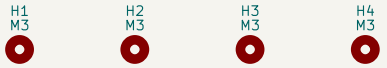




Aktiver 5V-Summer (12 mm), geschaltet ueber Q1.
D4 = Freilaufdiode

CO2-Ampel: gruen < 1000 ppm, gelb < 1500 ppm, rot >= 1500 ppm

Stromversorgung ueber die USB-Buchse des D1 mini.
5V = USB-Spannung, 3V3 = Regler auf dem D1 mini (max. ca. 500 mA)



ESP8266 + CO2-Sensor (I2C) + OLED + Ampel-LEDs

Sheet: /
File: CO2-Ampel.kicad_sch

Title: CO2-Ampel mit D1 mini

Size: A4	Date: 2026-09-29	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 1/1